Results

01 · CHI-SQUARE INDEPENDENCE

are two categorical variables related?

Table 1. Contingency (observed [expected] (row% / col%) std. residual)

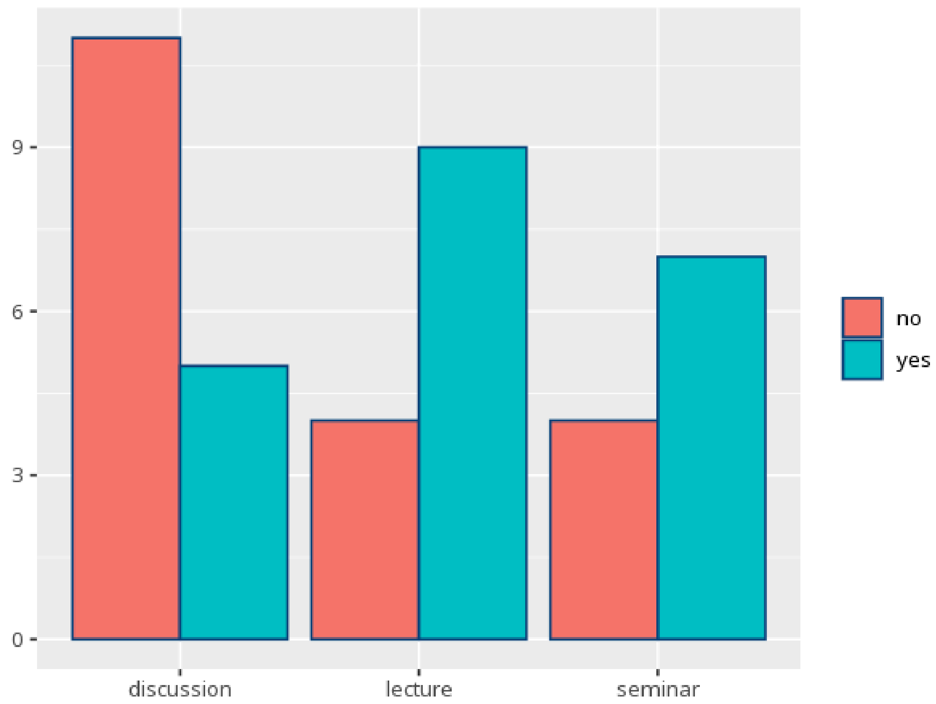
Row \ Column	no	yes	Total
discussion	11 [7.60] (68.8% / 57.9%) r=2.20	5 [8.40] (31.3% / 23.8%) r=−2.20	16
lecture	4 [6.17] (30.8% / 21.1%) r=−1.47	9 [6.83] (69.2% / 42.9%) r=1.47	13
seminar	4 [5.22] (36.4% / 21.1%) r=−0.87	7 [5.78] (63.6% / 33.3%) r=0.87	11
Total	19	21	40

Table 2. Chi-square test

χ^2	df	p	Cramér's V [95% CI]
4.90	2	.086	0.35 [0.00, 1.00]

the contingency table is r×c — columns expand to the number of categories in the column variable; each cell shows observed [expected] (row% / col%) and its standardized residual r (chisq.test()$stdres), with a warning (suggesting Fisher's exact) if expected counts are too small. A standardized residual beyond about ±1.96 flags a cell that significantly drives the association. For 2×2 tables chisq.test() applies Yates' continuity correction to the χ^2 by default (set correct=FALSE for the uncorrected χ^2); the standardized residuals are unaffected by the correction.

Figure. Cross-classification



How to read this test

Tests whether two categorical variables are associated. A p below alpha means they are related; Cramér's V gives the strength of association (0–1); each cell's standardized residual (r) beyond about ±1.96 flags a category combination that significantly drives the association. Valid only when expected counts are mostly ≥ 5 (the app flags this and suggests Fisher's exact) and each case appears once (raw counts, not percentages). Your significance threshold (α) is 0.05.

APA template: A chi-square test of independence gave $\chi^2(2, N=40)=4.90$, $p = .086$, $V=.35$ [.00, 1.00].